Saptarshi Mukhopadhyay

B.Tech CSE (AI/ML) student building wearable and embedded health tech
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About

I design and build physical systems that run machine learning on small devices. My projects connect sensors, analog circuits, microcontrollers and mobile apps, and aim at low-cost tools for healthcare and personal safety.

Projects

VoiceVault

Wearable silent speech device for laryngectomy patients

- Reads muscle (EMG) and vibration (piezoelectric) signals through an INA128 / TL072 / TL082 analog front end.
- Runs a TFLite Micro Siamese network on an ESP32-S3 and sends results over BLE to a Flutter app.
- Uses Hedera blockchain for voice-identity verification.

NeuroAI

Low-cost AI neuro-physiotherapy diagnostic system (in development)

- ESP32 platform with EMG/ECG (AD8232), PPG (MAX30102) and FSR pressure sensing.
- Python desktop app that produces patient reports.

Raksha

Women's safety Android app

- SOS system with GPS location and SMS alerts, designed to work offline first

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- Roadmap: shake-to-SOS, fake call, disguise mode, safe-walk timer, evidence capture.

Smart Voltage Stabilizer

Arduino / ESP32 for home use

Built the DC-side wiring on my own: voltage sensing, OLED output and relay control logic.

Other builds

Line-following robot, ESP32 voice assistant, Arduino radar system, and staircase LED automation with IR sensors and MOSFETs.

Skills

Embedded and hardware

ESP32, ESP32-S3, Arduino, EMG/ECG/PPG sensing, op-amp signal conditioning, BLE

AI / ML

Python

Apps

Python desktop apps

Security

Hands-on labs on TryHackMe

Experience

Google Student Ambassador

Google Gemini Student Ambassador Program 2026

Organize events, campaigns and workshops, and run student outreach on campus.

Hackathons and competitions

Hackathons and competitions

1st position, Mechnovate
Vivarta 9.0, Techno India University, with Blackbird Robotics; sponsored by GeeksforGeeks

PromptWars Virtual, Challenge 2 — Aug 2026
Google for Developers: submitted a verified Generative AI solution

InnoFusion 3.0 — 22–23 Aug 2026
30-hour offline software + hardware hackathon, IEM Kolkata

Code for Change 2.0 — 2026
Hackathon organized by CodeNEST, NSHM Knowledge Campus

Certifications

Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate — Oct 2025
Oracle University · valid until Oct 2027

Gemini Certified Educator — Nov 2025
Google for Education · valid until Nov 2028

Introduction to Data Science — Dec 2025
Cisco Networking Academy

Getting Started with Artificial Intelligence — Oct 2025
IBM SkillsBuild

Web Development, Cyber Security & Data Science (AI/ML) — 9–15 Sep 2025
6-day hands-on workshop, NSHM Knowledge Campus with Anovatech Institute of Crystalline Technology

Education

B.Tech, Computer Science & Engineering (AI/ML) Expected 2029

NSHM Knowledge Campus, Durgapur